

Flynn Duel
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BUSINESS CONTEXT AND PROBLEM

In this project, the goal is to predict how much a customer will spend on a specific product based on a combination of demographic and product-related information. The dataset comes from Walmart and includes a large number of past purchase records, each tied to an individual customer and product. We're treating this as a regression problem since the target variable, Purchase, is a continuous number that shows how much money was spent in a single transaction. If we can build a model that predicts that amount accurately, it could help Walmart better understand customer behavior and improve how they plan promotions, stock inventory, or recommend products.

From a business perspective, being able to predict purchase behavior is valuable. If Walmart knows which customers are likely to spend more on certain types of products, they can use that information to plan more targeted marketing, better product recommendations, and smarter pricing. They could also make better inventory decisions across different regions. Since Walmart operates at such a large scale, even small improvements in prediction accuracy could lead to significant cost savings and higher revenue.

From a broader perspective, retail companies often struggle with overstocking or understocking certain product lines, leading to lost revenue or excess inventory costs. By analyzing historical data and building a predictive model, Walmart can improve planning and execution across the value chain. Even small improvements in accuracy at Walmart's scale can translate into significant operational and financial gains.

Variable Reference Table

VARIABLE NAME	DESCRIPTION
User_ID	Unique identifier for each customer
Product_ID	Unique identifier for each product
Gender	Customer gender (Male/Female)
Age	Age group of the customer (e.g., '26-35')
Occupation	Occupational category (numeric code)
City_Category	City category based on location tier (A, B, or C) A = City B = Mid-tier C = Rural
Stay_In_Current_Years	Years spent in the current city by the customer
Marital_Status	Customer marital status (0 = Single, 1 = Married)
Product_Category	Product category ID (encoded)
Purchase	Dollar amount spent on the product (TARGET VARIABLE)

DATA CLEANING SUMMARY

The dataset was already in pretty good shape, with no missing values to deal with. That made the initial cleaning process easier, but there were still a few small changes that needed to be made to prepare the data for modeling. First, the Stay_In_Current_City_Years column had a value labeled “4+”, which I changed to just 4 so the entire column could be treated as a numeric

variable. I also converted the relevant columns into factors, including Gender, Age, City_Category, Occupation, and Marital_Status. These are all categorical variables, and treating them as factors helps the model understand how different groups of customers behave. Lastly, I removed User_ID and Product_ID from the modeling dataset since they are just identifiers and don't carry any predictive value.

TARGET VARIABLE: PURCHASE

The variable we are trying to predict in this project is Purchase, which represents how much money a customer spent on a given product. Since it's a continuous number and not a category, this confirms that the problem we are solving is a regression problem. I looked at the distribution of purchase amounts by plotting a histogram, and the results showed a somewhat uneven but still right-leaning shape. Most purchases were clustered between \$2,000 and \$10,000, with a few noticeable spikes at regular intervals. Some larger transactions stretched past \$20,000, which could reflect big-ticket items or bulk purchases. This confirmed that Purchase is a continuous variable with a wide range, making it appropriate for regression modeling.

VISUALIZATIONS

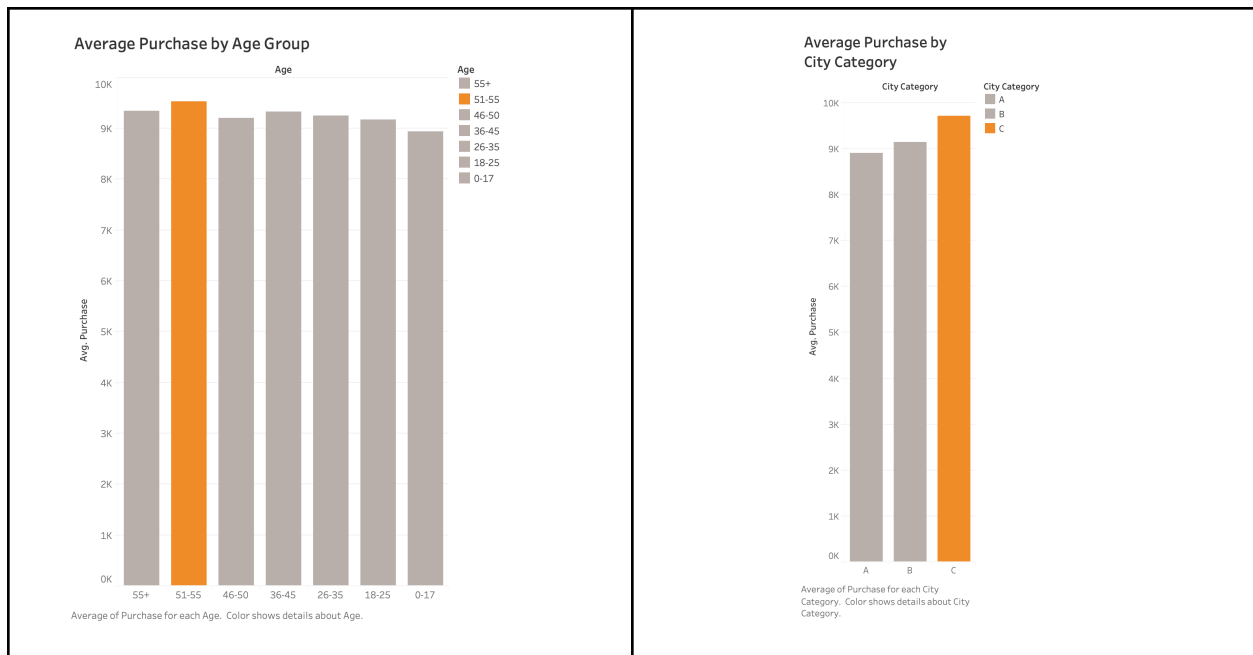


Figure 1: *Customers aged 51–55 spend the most per transaction*

Figure 2: *City Category C (rural) leads in average purchase amount*

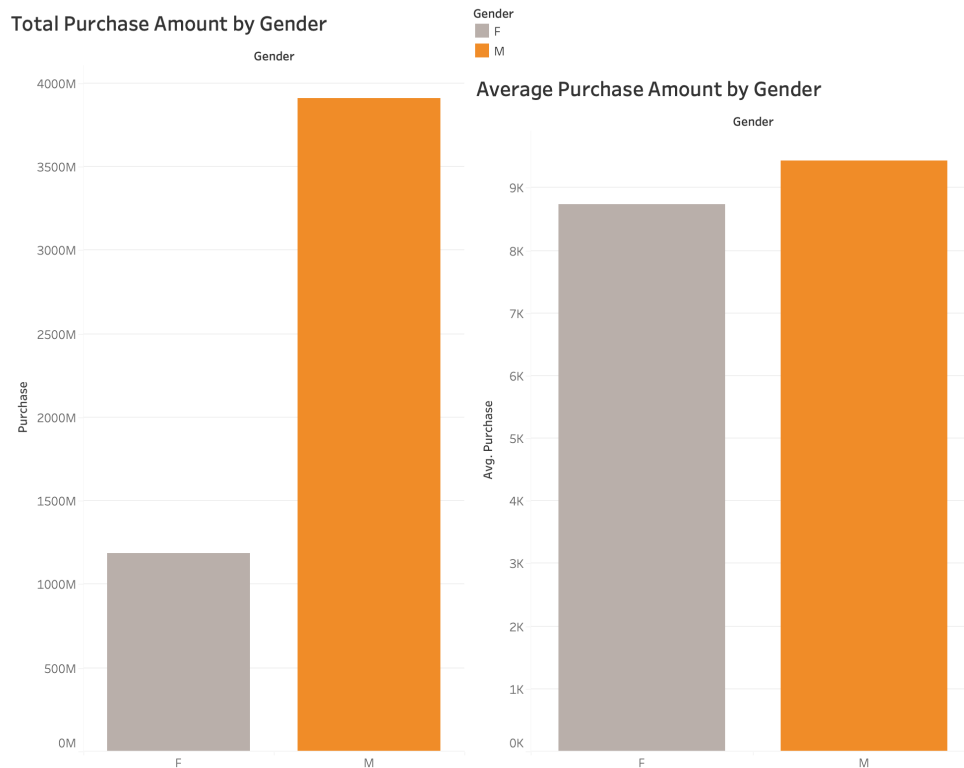


Figure 3: *Although male customers account for a larger share of total purchases, both genders demonstrate similar average spending per transaction.*

VISUAL INSIGHTS

To get a better understanding of the data, I created a few visualizations that look at customer behavior based on different features. One chart compares average purchase amount by age group, and it shows that people in the 51 to 55 range tend to spend the most per transaction. This group might be a key target for higher-end product recommendations or promotions. Another chart breaks down average purchases by city category, and it turns out that customers in Category C cities spent the most on average, which could suggest that rural cities are a strong market segment. Lastly, I made a small dashboard that compares total spending by gender to average spending per transaction. Men contributed significantly more to total revenue overall, but the average amount spent per transaction was basically the same for both genders. That tells us that the volume of purchases might be driving the total difference, not spending habits per

purchase.

FEATURE ENGINEERING AND DATA TRANSFORMATION

After cleaning the data, I made two different versions of the dataset so I could use them with the two models I plan to run. The first version was for the Random Forest model. Since Random Forest can work directly with categorical variables, I kept the factor variables as they were. This included columns like gender, age group, occupation, and city category. The second version was for the Elastic Net regression model, which requires all the input variables to be numeric. To prepare for that, I used the `dummyVars()` function from the `caret` package. This turned each category in the factor variables into its own column with values of 0 or 1. This process is called dummy encoding and helps the model treat each category separately. After the encoding, I added the Purchase column back in so the model could learn from it. Making both versions of the data helped make sure each model had what it needed to run properly without changing the meaning of the variables.

MODEL SELECTION AND TRAINING

For this project, I chose two models to train and compare: Random Forest and Elastic Net regression. These models were selected based on the type of problem we are solving, which is predicting a continuous outcome variable (Purchase) based on a mix of categorical and numeric features. Random Forest was a good choice because it can handle a lot of different variable types, does not require scaling, and can pick up non-linear relationships in the data. It is also helpful for business analysis because it gives variable importance scores, which show which features have the biggest impact on predictions. Elastic Net regression was chosen because it is a strong model for creating a simpler, more interpretable structure. It automatically shrinks less important coefficients to zero, meaning it selects only the most useful variables while leaving out noise. This is especially useful in a dataset like this one, where dummy encoding of categorical variables creates a lot of predictors.

For Random Forest, I used 10-fold cross-validation to tune the `mtry` parameter, which is the number of predictors considered at each split. I tested values from a selected range (4, 6, 8, and 10) rather than testing all possible values, which helped the model run more efficiently while

still producing strong results. I also kept the number of trees at 500 to maintain stable model performance. For Elastic Net, I used 5-fold cross-validation and tuned the lambda penalty value over a wide range to find the best fit. Both models were trained using the `train()` function from the `caret` package, which handled the cross-validation and hyperparameter search. This process helped make sure the models were trained consistently, tuned properly, and were less likely to overfit to the training data.

PERFORMANCE METRICS

<u>MODEL</u>	<u>RMSE</u>	<u>R²</u>	<u>MAE</u>
Random Forest	3820.05	0.46	2862.18
Elastic Net	4703.95	0.13	3603.57

After training and testing both models, I evaluated their performance using RMSE, R^2 , and MAE on the test set. The Random Forest model achieved an RMSE of 3820.05, an R^2 of 0.46, and an MAE of 2862.18. This means that, on average, the Random Forest model's predictions were within about \$2,862 of the true purchase amount, and that about 46% of the variation in customer spending could be explained by the model. The Elastic Net model performed significantly worse, with an RMSE of 4703.95, an R^2 of just 0.13, and an MAE of 3603.57. While Elastic Net was helpful for feature selection and reducing model complexity, it was not very good at accurately predicting purchase amounts. Based on these results, Random Forest would be the stronger choice for making real-world predictions, even though there is still room for improvement in overall model fit.

BUSINESS FINDINGS

After training and evaluating both models, the Random Forest model was able to find some meaningful patterns in the data, even though it was working with limited information. Customers aged 51–55 were identified as one of the groups most likely to spend more per transaction. This finding is important because it suggests that older customers could be a valuable target for promotions, especially for higher-priced or luxury products. The model also

found that customers from Category C cities (rural areas) tended to spend slightly more on average. Even though rural markets are sometimes assumed to be less profitable, this result suggests that Category C cities may represent an underutilized opportunity for Walmart's marketing and inventory planning.

Occupation and Marital Status also showed up as moderately important predictors, although their impact was not as strong as Age and City Category. These features could still help Walmart fine-tune its targeting, for example by distinguishing marketing strategies between single and married customers, or between different types of jobs that might influence disposable income. Elastic Net was not very successful at making accurate predictions, but it did help confirm that not every variable contributed equally to purchase behavior. Features like Gender, which might have been expected to play a bigger role, ended up having relatively little predictive power when compared to other factors. Overall, while the models were not perfect, they still uncovered useful insights that Walmart could apply to focus its marketing and operations more effectively.

RECCOMENDATIONS

Based on the analysis, Walmart should focus more heavily on older customers, especially those in the 51–55 age range, who tend to spend more per transaction. Promotions for higher-end or larger-ticket products could be specifically targeted toward this demographic group. Walmart should also look closer at Category C cities (rural areas) because customers there showed higher average spending than expected. Inventory and marketing plans could be adjusted to better meet demand in these regions rather than assuming rural areas are lower priority.

From a modeling standpoint, Walmart would benefit from gathering more detailed data on customers, especially related to income, promotion exposure, loyalty program participation, and product preferences. Right now, the available data only scratches the surface of what might be influencing purchase behavior. Finally, if Walmart needs to actually predict purchase amounts today based on this dataset, Random Forest would be the better model to deploy because it provided more reliable predictions compared to Elastic Net.

LIMITATIONS

One major limitation of this project is the type and amount of data that was available. The dataset mainly included basic demographic features like age group, marital status, city category, and occupation, but it left out a lot of other important factors that could have had a strong impact on purchasing behavior. For example, we didn't have any information about customer income, education level, loyalty program enrollment, past buying habits, or whether purchases were influenced by promotions or discounts. These missing pieces are probably a big reason why even the best model, Random Forest, was only able to explain about 46% of the variation in spending.

Another limitation is that the dataset treated every purchase event separately without providing a full history for each customer. In reality, shopping patterns tend to be influenced by cumulative habits, not just individual transactions. Since we couldn't model behavior over time, our predictions were based only on snapshot-level features. Also, Elastic Net showed that many of the available variables were not strongly predictive on their own. This makes sense, but it also points back to a bigger issue: it's hard to predict complex behavior like spending with limited and fairly surface-level data. These limitations don't mean the project wasn't useful, but they should definitely be considered when thinking about how far the results can be applied.

FUTURE WORK

One of the biggest opportunities for improving this project would be to collect a richer set of features that go beyond the basic demographic information we had available. Right now, most of the features are surface-level, like age group, city type, marital status, and occupation. While those can tell part of the story, a lot of important factors that likely influence how much someone spends were missing. For example, having information on customer income, education level, loyalty program membership, or even responsiveness to past promotions could give the models a lot more predictive power. It would also be really helpful to track how customers spend over time rather than just looking at individual transactions. Most people's buying habits are shaped by past behavior, and having that kind of history would let models learn those patterns instead of trying to guess from a single purchase.

If the dataset were expanded and included better quality features, it could open the door to trying more complex models. One idea would be to experiment with Neural Networks. Neural Networks are known for being able to pick up complicated relationships between variables, especially when there are interactions that simpler models might miss. However, they usually require a lot more data and careful tuning to perform well, which is why we didn't use them in this project. Even without switching models, just giving the existing Random Forest and Elastic Net models better inputs would probably lead to stronger results. The focus, moving forwards, should be on building a more complete and detailed dataset that reflects the real factors behind customer spending.